

Mathom Johnson

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EDUCATION

University of Wisconsin—Madison

Madison, WI

B.S. Computer Science, B.S. Data Science — GPA: 3.92/4.0

Sep. 2023 – May 2027

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Linear Algebra, Discrete Math, Data Science Modeling, Big Data Systems, Time Series Analysis, Intro to Artificial Intelligence

EXPERIENCE

Embedded Systems Engineering Intern

Sep. 2026 – Present

Plexus, Corp.

Neenah, WI

- Completing a six-week C++ bootcamp covering embedded systems development, object-oriented programming, and UML in preparation for assignment to an embedded systems project

Software Engineer Intern

Jun. 2026 – Aug. 2026

U.S. Bank

Minneapolis, MN

- Investigated a PostgreSQL connection pooling incident caused by mismatched client/server pooling configs, leading to zombie connection buildup and app outages; developed standardized pooling modules to prevent recurrence across app dev teams
- Built a Python capacity monitoring tool that analyzed usage across 1,000+ Fabric workspaces to identify inefficient workloads, orchestrating scheduled email alerts to workload owners via a Fabric Data Pipeline

Co-Founder & Engineer

Oct. 2025 – Present

Praxora Education, Inc.

Madison, WI

- Co-founded and serve on the board of an incorporated ed-tech startup that has raised \$30,000 in pre-seed funding
- Architected and shipped the backend for an AI-powered OSCE practice platform (FastAPI, Supabase, Railway), including OTP authentication, row-level security policies, Redis-based rate limiting, and Stripe integration
- Built a timezone-aware scheduled email system using precomputed UTC timestamps and idempotency keys to guarantee exactly-once delivery of weekly practice reminders across distributed cron workers

Undergraduate Researcher, Embedded Systems

Sep. 2025 – May 2026

Morgridge Institute for Research

Madison, WI

- Implemented an emergency shutoff control that disables high-voltage outputs across all three cathode heating channels in a research-grade 3D metal printer
- Built and deployed a remote monitoring web application that streams live telemetry from the 3D metal printer, letting researchers track interlocks, vacuum pressure, and cathode temperatures remotely without staffing the lab
- Designed the supporting PostgreSQL pipeline, instrumenting the dashboard to emit 3-second telemetry and using pg_partman partitioning with pg_cron aggregation to compress logs into minute averages, reducing storage 20x

Undergraduate Researcher, Data Analysis

Feb. 2025 – May 2026

Ehrlich Lab, University of Wisconsin—Madison

Madison, WI

- Deployed an open-source machine learning model as a CLI tool to classify tail bouts from tracking data, increasing data processing speed by over 2x and enabling non-technical researchers to run classifications independently
- Designed and implemented experimental hardware using Arduino and IoT components to develop a new larval zebrafish behavioral assay, demonstrating strong problem-solving skills from concept to execution

Software Development Intern

May 2025 – Aug. 2025

WEC Energy Group

Milwaukee, WI

- Developed and deployed a data pipeline that manipulates, aggregates, and stores over 2,200 data objects daily, working with a cross-functional agile team
- Built unit and integration test suites raising code coverage to 92%, and refactored core functions into modular components to improve reusability and testability

TECHNICAL SKILLS

Languages: Java, Python, C, C#, JavaScript, TypeScript, HTML/CSS, R, PowerShell, MATLAB, SQL, Assembly

Frameworks & Tools: React.js, Node.js, Ollama, ChromaDB, Flask, Django, .NET, Git, GCP, Redis, Supabase

Hobbies & Interests: Paddleboarding, Cursor, Gregory Alan Isakov, Soccer, Tiny Houses, Running